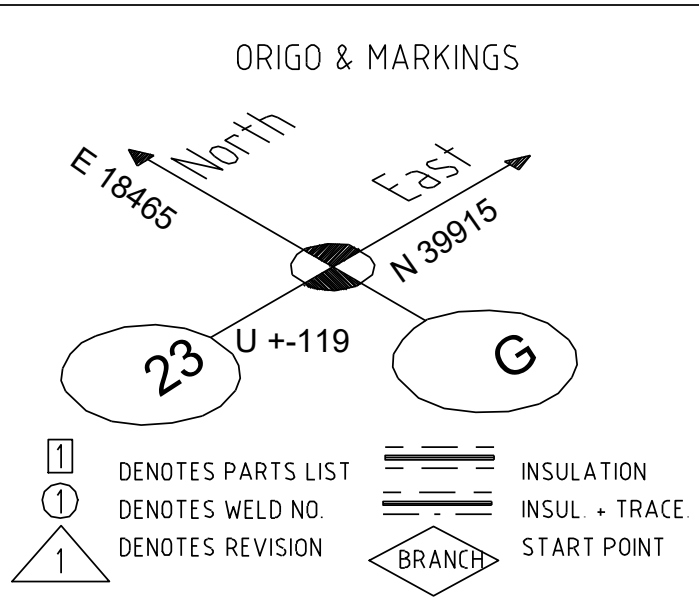




**NOTE:**

Site weldings to be added by contractor.

Buildings construction and equipment erection tolerances to be checked by contractor.

Pipe levels and dimensions are according to pipe center line.

Isometric drawings for sizes DN 15-25 are approximate.

Piping contractor is responsible for final routing and supporting.

|                         |            |                       |                     |
|-------------------------|------------|-----------------------|---------------------|
|                         |            | Hazardous CLP content |                     |
| Directive               | 2014/68/EU | Pipe Class            | /119440-VE16C1C-R00 |
| Design press. (min)     | -1bar      | PED Category          | I                   |
| Design temp. (min)      | -          | Media                 |                     |
| Max. allow. press. (PS) | -          | Standard              |                     |
| Operating press.        | 5.9bar     | P&ID                  | RAU8F00303          |
| Operating temp.         | 181degC    | Design press.         | 6bar                |
| PED Module              | -          | Design temp.          | 220degC             |
| PED Group               | -          | Test press.           | -                   |
| Wind loads              | -          | Pipe list doc. ID     | RAUAF04300          |
| External loads          | -          | Valve list doc. ID    | RAUAF04301          |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                       |                             |  |               |               |                |      |          |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------|--|---------------|---------------|----------------|------|----------|--|
| <p>For installation tolerances dimensions to be checked by pipe contractor at site.<br/> Degree of prefabrication to be decided by contractor.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                       |                             |  |               |               |                |      |          |  |
| Rev                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Revision Note                                                                         | Created by                  |  | Checked by    |               | Approved by    |      | Date     |  |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                       | J. Lakka                    |  | T. Saarinen   |               | M. Kankkunen   |      | 5/5/2016 |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Material                                                                              | Equipment Type              |  |               |               | Equipment ID   |      |          |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Tervasaari, Valkeakoski                                                               | Höyry- ja lauhdejärjestelmä |  |               |               |                |      |          |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Project Key                                                                           | Project Name                |  |               |               | Project Number |      | Work     |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | TERPMS_N52654                                                                         | Teräksinen Safe Future      |  |               |               | N52654         |      |          |  |
| Projection Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  | General Tolerances          |  | ISO 2768-mK-E | ISO 2768-mK-E | Scale          | Size | Weight   |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ISO 2553                                                                              | ISO 13 920-BF               |  | ISO 2768-mK-E | ISO 2768-mK-E | -              | A1   |          |  |
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| <p>Title</p> <p>50173-HMP-80-16C1B</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                       |                             |  |               |               |                |      |          |  |
| Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                       | Document ID                 |  |               |               |                |      | Sheet    |  |
| CERTIFIED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                             |  |               |               |                |      | 1 / 1    |  |
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                       |                             |  |               |               |                |      | Revision |  |
| Internal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                       |                             |  |               |               |                |      | 0        |  |